NEET UG PHYSICS — MECHANICS

Laws of Motion

Syllabus Context: Dynamics of particles and systems, inertia, Newton's three laws of motion, linear momentum and impulse, law of conservation of linear momentum, concurrent forces and equilibrium, static, kinetic, and rolling friction, dynamics of uniform circular motion (level and banked roads), and free-body diagram problem-solving techniques.

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1. Learning Objectives & Chapter Overview

In kinematics, we describe the motion of a particle quantitatively in terms of velocity (for uniform motion) and acceleration (for non-uniform motion) without asking what causes or governs that motion. In this tutorial, we address the fundamental question of dynamics: **What governs the motion of bodies?**

Key Learning Objectives for NEET UG

- Understand why Aristotle's view of motion fails and how Galileo discovered the law of inertia using inclined planes.
- Apply **Newton's First Law of Motion** to identify states of zero acceleration and determine unknown forces in equilibrium.
- Master **Newton's Second Law of Motion** ($\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = m\mathbf{a}$), component wise resolution, local nature of force, and the concept of **Impulse**.
- Analyze mutual interactions using **Newton's Third Law of Motion** and avoid common misconceptions regarding action–reaction pairs.
- Apply the **Law of Conservation of Momentum** to isolated systems, gun-bullet recoil, nuclear disintegration, and two-body collisions.
- Solve problems involving **equilibrium of concurrent forces**, tension in strings, pulleys, and springs.
- Distinguish between **static, limiting, kinetic, and rolling friction**, and calculate accelerations and contact forces on horizontal and inclined planes.
- Analyze the dynamics of **uniform circular motion** on level and banked circular tracks, including optimum speed and maximum safe speed.
- Construct accurate **Free-Body Diagrams (FBDs)** for multi-body assemblies.

2. Core Concepts & Detailed Explanations

2.1 Introduction: Force and External Agencies

Common experience tells us that to move a stationary football, someone must kick it. To throw a stone upwards, one must give it an upward push. A breeze swings the branches of a tree, and a flowing river moves a boat. Similarly, an external agency is required to retard or stop a moving body—such as applying a force against the direction of a ball rolling down an inclined plane.

- **Contact Forces:** In many everyday situations, the external agency (hands, wind, water stream) is in direct physical contact with the object.
- **Non-Contact Forces (Action at a Distance):** Contact is not always necessary. A stone released from the top of a building accelerates downward due to the Earth's gravitational pull. A bar magnet attracts an iron nail from a distance. Thus, gravitational, electrical, and magnetic agencies can exert forces from a distance without an intervening medium.

In short, a force is required to put a stationary body in motion or to stop a moving body, and some external agency is needed to provide this force.

2.2 Aristotle's Fallacy vs. Galileo's Law of Inertia

What if a body is moving uniformly (for example, a skater moving straight with constant speed on a horizontal ice slab)? Is an external force required to keep a body in uniform motion?

The Greek thinker **Aristotle (384 B.C. – 322 B.C.)** held the view that if a body is moving, something external is continuously required to keep it moving. For instance, he believed an arrow shot from a bow keeps flying because the air behind the arrow keeps pushing it. Stated concisely, the **Aristotelian law of motion** claimed: *An external force is required to keep a body in motion.*

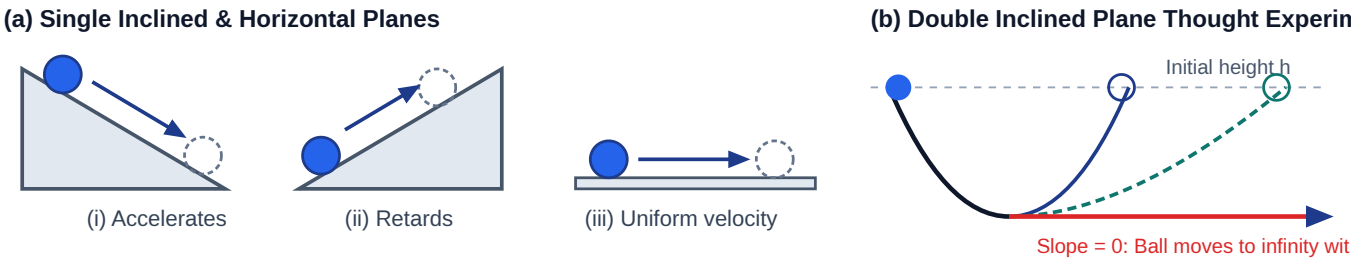
Why Aristotle's View is Flawed: A child dragging a non-electric toy car with a string must pull continuously to keep it going; if the string is released, the car comes to rest. Aristotle coded this everyday terrestrial experience into a basic law,

overlooking the fact that the car stops because the external force of **friction** exerted by the floor opposes its motion. When the child pulls the car at constant velocity, the applied force by the child cancels the frictional force by the floor, resulting in **zero net external force**. If friction and viscous drag were completely absent, no external force would be needed to maintain uniform motion.

Galileo's Inclined Plane Experiments

In the seventeenth century, **Galileo Galilei** imagined an idealized frictionless world and studied the motion of objects on inclined planes:

1. Objects moving **down** an inclined plane **accelerate**.
2. Objects moving **up** an inclined plane **retard** (decelerate).
3. Motion on a **horizontal plane** is an intermediate situation: on a frictionless horizontal plane, an object must experience neither acceleration nor retardation, moving with **constant velocity**.



Galileo's Equivalence Principle & Inertia:

- State of rest (v = 0) and state of uniform linear motion (v = constant) are completely equivalent: both have F_net = 0.
- Inertia ("resistance to change"): The inherent property of a body by which it resists any change in its state of rest or uniform moti

Tutorial Diagram 1: Galileo's observations on single and double inclined planes leading to the Law of Inertia.

In his second experiment using a **double inclined plane**, a ball released from rest on one smooth plane rolls down and climbs up the second plane to nearly the same initial height (and in the ideal frictionless limit, to the exact same height). When the slope of the second plane is decreased, the ball still reaches the same vertical height but travels a longer distance. In the limiting case where the slope of the second plane is zero (horizontal), the ball travels an infinite distance—its motion never ceases.

Historical Note: Ideas on Motion in Ancient Indian Science

Ancient Indian thinkers developed an elaborate framework of ideas on motion. In the *Vaisesika* school, force (the cause of motion) was classified into several types: force due to continuous pressure (*nodan*, such as wind on a sailing vessel), impact (*abhighat*, such as a potter's rod striking a wheel), persistent tendency (*sanskara*) to move in a straight line (*vega*) or restore shape in an elastic body, and transmitted force via a string or rod.

The notion of **vega** (tendency to move in a straight line, opposed by contact with objects and the atmosphere) comes closest to the concept of inertia and friction. Thinkers also recognized that translational, rotational (*bhraman*), and vibrational (*spandan*) motions of an extended body (such as a falling leaf, *patan*) arise from the translational displacements of its constituent particles. In 1150 A.D., **Bhaskara** introduced 'instantaneous motion' (*tatkaliki gati*), anticipating instantaneous velocity in differential calculus.

2.3 Newton's First Law of Motion

Building upon Galileo's law of inertia, Isaac Newton formulated the foundation of classical mechanics.

Newton's First Law of Motion: Every body continues to be in its state of rest or of uniform motion in a straight line unless compelled by some external force to act otherwise.

Equivalent Operational Statement: If the net external force on a body is zero ($\mathbf{F}_{\text{net}} = \mathbf{0}$), its acceleration is zero ($\mathbf{a} = \mathbf{0}$). Acceleration can be non-zero only if there is a net external force acting on the body.

In practice, we apply Newton's First Law in two ways:

1. **Known Zero Net Force $\Rightarrow$ Zero Acceleration:** A spaceship out in interstellar space, far from all stars and planets and with all its rockets turned off, experiences zero net external force. Therefore, its acceleration must be zero; if it is moving, it continues moving with constant velocity.
2. **Observed Zero Acceleration $\Rightarrow$ Zero Net Force:** On Earth, gravity and friction are pervasive. If an object on Earth is observed to be at rest or in uniform linear motion ($\mathbf{a} = \mathbf{0}$), it is not because no forces act on it, but because all external forces cancel out to give zero net external force.

Crucial Conceptual Distinction: Book on a Table & Moving Car

- **Book at Rest on a Horizontal Table:** Two external forces act on the book: its weight W acting downward (due to Earth's gravity) and the upward normal force R exerted by the table (a self-adjusting contact force).
  Incorrect Reasoning: "Since $W = R$, the forces cancel and therefore the book is at rest."
  Correct Physical Reasoning: "We observe the book to be at rest ($\mathbf{a} = \mathbf{0}$). By Newton's First Law, the net external force must be zero, which implies that the normal force R adjusts itself to be equal and opposite to W."
- **Car Starting from Rest:** When a car accelerates during pick-up, the acceleration of the car as a whole cannot be caused by any internal force of the engine. The only external horizontal force along the road is the **force of friction** between the tyres and the road, which accelerates the car forward. When the car moves at constant velocity, the net external force on it is zero.
- **Passenger in a Bus (Inertia of Rest and Motion):** Because the human body is deformable (not strictly rigid), when a stationary bus starts suddenly, friction between the floor and the passenger's feet accelerates the feet forward with the bus, while the upper body remains behind due to inertia—throwing the passenger backward relative to the bus until muscular forces restore posture. Conversely, when a moving bus stops suddenly, friction stops the feet while the upper body continues moving forward due to inertia, throwing the passenger forward.

2.4 Linear Momentum and Newton's Second Law of Motion

Momentum ($\mathbf{p}$) of a body is defined as the product of its mass m and velocity $\mathbf{v}$:

$$\mathbf{p} = m\mathbf{v}$$

Momentum is a vector quantity directed along the velocity $\mathbf{v}$. Its SI unit is $\mathrm{kg\,m\,s^{-1}}$ (or $\mathrm{N\,s}$) and its dimensional formula is $[\mathrm{M\,L\,T^{-1}}]$.

Physical Clues Leading to the Second Law

- **Role of Mass and Speed:** Pushing a heavy truck from rest to a certain speed in a given time requires a much greater force than pushing a light car. Catching a heavy stone dropped from a building is harder than catching a light stone. A bullet fired at high speed pierces tissue, whereas the same bullet at moderate speed causes little harm. Thus, the product $m\mathbf{v}$ is the crucial physical variable of motion.
- **Rate of Change of Momentum (Catching a Cricket Ball):** A seasoned cricketer draws his hands backward while catching a fast cricket ball, increasing the time interval Δt taken to stop the ball and thereby reducing the required force. A novice keeps his hands fixed, stopping the ball in a tiny time interval, which requires a huge force that hurts his hands. Thus, force depends not just on $\Delta\mathbf{p}$, but on the *rate* $\frac{\Delta\mathbf{p}}{\Delta t}$.
- **Vector Change in Circular Motion:** When a stone is whirled with uniform speed in a horizontal circle by a string, the magnitude of momentum $p = mv$ is constant, but its direction changes continuously. A force (provided by the hand through the string) is required to change the direction of the momentum vector. Greater speed or smaller radius requires a greater force.

Newton's Second Law of Motion: The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts.

Mathematically, if a force $\mathbf{F}$ acts for a time interval Δt changing the momentum by $\Delta \mathbf{p}$, then in the limit $\Delta t \to 0$:

$$\mathbf{F} = k\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}$$

Choosing the definition of the unit of force such that the proportionality constant $k = 1$, and considering a body of fixed mass m where $\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \frac{\mathrm{d}(m\mathbf{v})}{\mathrm{d}t} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = m\mathbf{a}$, we obtain:

$$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = m\mathbf{a}$$

SI Unit of Force (newton, N): One newton ($1\,\mathrm{N}$) is the force that causes an acceleration of $1\,\mathrm{m\,s^{-2}}$ in a body of mass $1\,\mathrm{kg}$. Thus, $1\,\mathrm{N} = 1\,\mathrm{kg\,m\,s^{-2}}$.

Four Essential Features of Newton's Second Law

1. **Consistency with First Law:** Setting $\mathbf{F} = \mathbf{0}$ yields $\mathbf{a} = \mathbf{0}$, which directly recovers Newton's First Law.
2. **Vector Law (Independent Orthogonal Components):** The vector equation $\mathbf{F} = m\mathbf{a}$ is equivalent to three scalar equations along Cartesian axes:
   $$F_x = \frac{\mathrm{d}p_x}{\mathrm{d}t} = ma_x, \qquad F_y = \frac{\mathrm{d}p_y}{\mathrm{d}t} = ma_y, \qquad F_z = \frac{\mathrm{d}p_z}{\mathrm{d}t} = ma_z$$
   A force acting at an angle to the velocity changes *only* the component of velocity along the direction of the force; the component of velocity perpendicular (normal) to the force remains completely unchanged (e.g., horizontal velocity of a projectile under vertical gravity).
3. **Applicability to a System of Particles:** For a rigid body or system of particles, $\mathbf{F}$ represents the **net external force** on the system, and $\mathbf{a}$ represents the acceleration of the centre of mass of the system. Internal forces between parts of the system cancel out and must not be included in $\mathbf{F}$.
4. **Local Relation in Space and Time (No Memory of Past Acceleration):** Force $\mathbf{F}$ at a specific point in space at a specific instant t determines the acceleration $\mathbf{a}$ at that exact point and instant. Acceleration carries **no memory** of the prior history of motion. For example, the moment a stone is released from the window of an accelerating train (neglecting air resistance), the horizontal force on the stone becomes zero immediately; its horizontal acceleration is zero, and it experiences only the downward acceleration g due to gravity (even though it retains the horizontal velocity it had at the instant of release).

Impulse and Impulsive Forces

When a very large force acts for a very short time duration (such as a bat hitting a ball or a ball bouncing off a rigid wall) to produce a finite, measurable change in momentum, we define **Impulse ($\mathbf{J}$ or $\mathbf{I}$)** as the product of force and time duration:

$$\text{Impulse} = \mathbf{F} \times \Delta t = \Delta \mathbf{p} = \mathbf{p}_{\text{final}} - \mathbf{p}_{\text{initial}}$$

An **impulsive force** is not a conceptually new category of force in Newtonian mechanics; it is simply an ordinary force that is very large and acts over a short time interval so that no appreciable change in position occurs during the impact.

2.5 Newton's Third Law of Motion

In Newtonian mechanics, forces never occur singly in nature; every external force on a body arises from its mutual interaction with another body.

Newton's Third Law of Motion: To every action, there is always an equal and opposite reaction.

Precise Physical Formulation: Forces always occur in pairs. The force exerted on body A by body B ($\mathbf{F}_{AB}$) is equal in magnitude and opposite in direction to the force exerted on body B by body A ($\mathbf{F}_{BA}$):

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

Three Critical Rules for Action–Reaction Pairs

1. **Action and Reaction Mean Force:** Both terms simply refer to forces arising from mutual interaction. Even when bodies are not in contact—such as the Earth pulling a falling stone downward—the stone simultaneously exerts an equal and opposite upward gravitational pull on the Earth (we do not notice the Earth's acceleration because its mass is enormous).
2. **Simultaneous Nature (No Cause–Effect Delay):** Action does not precede or "cause" reaction. Both $\mathbf{F}_{AB}$ and $\mathbf{F}_{BA}$ act at the exact same instant. Either force may be called action and the other reaction.
3. **Act on Two Different Bodies:** $\mathbf{F}_{AB}$ acts on body A, while $\mathbf{F}_{BA}$ acts on body B. Because they act on different bodies, **action and reaction forces can never cancel each other when analyzing the motion of a single body**. However, if we consider the combined system $(A + B)$ as a whole, $\mathbf{F}_{AB}$ and $\mathbf{F}_{BA}$ become internal forces and sum to zero.

2.6 Conservation of Linear Momentum

Combining Newton's Second and Third Laws yields a fundamental conservation law. Consider two interacting bodies A and B in an **isolated system** (zero net external force) that collide or interact for a common contact time interval Δt.

By the Second Law, the momentum changes are $\mathbf{F}_{AB}\Delta t = \mathbf{p}'_A - \mathbf{p}_A$ and $\mathbf{F}_{BA}\Delta t = \mathbf{p}'_B - \mathbf{p}_B$. Since $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$ by the Third Law:

$$\mathbf{p}'_A - \mathbf{p}_A = -(\mathbf{p}'_B - \mathbf{p}_B) \implies \mathbf{p}'_A + \mathbf{p}'_B = \mathbf{p}_A + \mathbf{p}_B$$

Law of Conservation of Momentum: The total linear momentum of an isolated system of interacting particles is conserved (remains constant in both magnitude and direction).

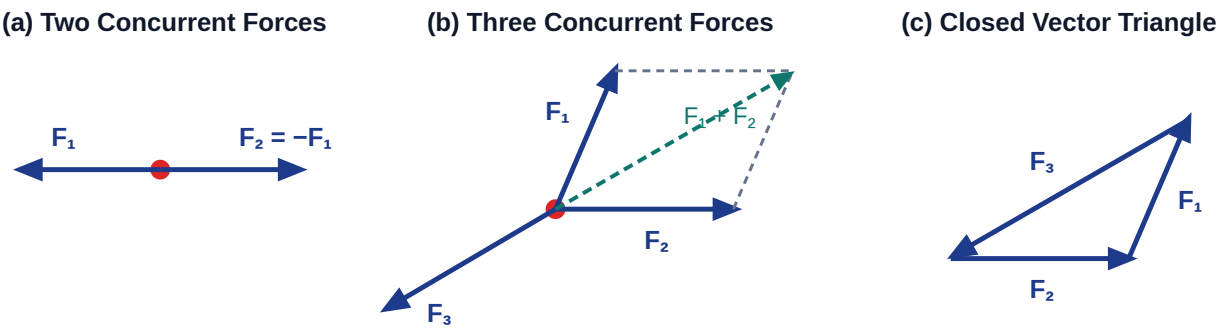
- **Gun and Bullet System:** Initially both gun and bullet are at rest ($\mathbf{p}_{\text{total}} = \mathbf{0}$). Upon firing, if the bullet acquires forward momentum $\mathbf{p}_b$, the gun acquires recoil momentum $\mathbf{p}_g$ such that $\mathbf{p}_b + \mathbf{p}_g = \mathbf{0}$, or $\mathbf{p}_g = -\mathbf{p}_b$.
- **Universal Validity in Collisions:** Momentum conservation holds strictly in *both* elastic and inelastic collisions of an isolated system.

2.7 Equilibrium of a Particle

In mechanics, a particle is in **translational equilibrium** when the net external force acting on it is zero ($\sum \mathbf{F} = \mathbf{0}$), meaning the particle is either at rest or moving with uniform velocity. (For an extended rigid body, complete equilibrium additionally requires rotational equilibrium, i.e., zero net external torque).

- **Two Forces:** Equilibrium requires $\mathbf{F}_1 = -\mathbf{F}_2$ (equal in magnitude, opposite in direction).
- **Three Concurrent Forces:** Equilibrium requires $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$. Graphically, the resultant of any two forces (by the parallelogram law) is equal and opposite to the third force, or the three force vectors taken in the same cyclical order form a **closed triangle**. In general, n concurrent forces are in equilibrium if they form a closed n-sided polygon taken in the same sense.
- **Component Equations for Equilibrium:**

$$\sum F_x = F_{1x} + F_{2x} + F_{3x} = 0, \qquad \sum F_y = F_{1y} + F_{2y} + F_{3y} = 0, \qquad \sum F_z = F_{1z} + F_{2z} + F_{3z} = 0$$



Tutorial Diagram 2: Equilibrium of a particle under concurrent forces and the closed triangle representation.

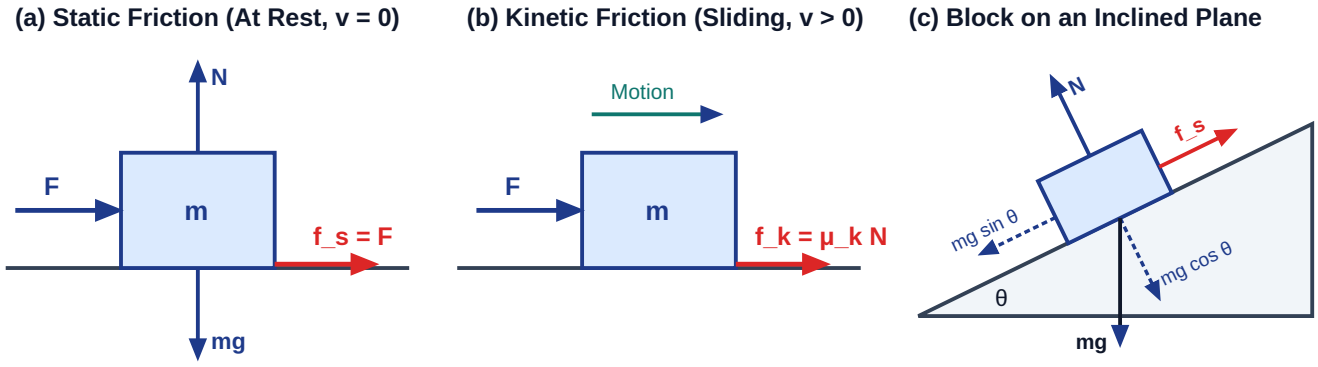
2.8 Common Forces in Mechanics

Of the four fundamental forces in nature (gravitational, electromagnetic, strong nuclear, and weak nuclear), only **gravitational** and **electrical** forces are relevant to macroscopic mechanics. While gravity is a pervasive non-contact force, all common mechanical **contact forces** ultimately arise at the microscopic level from electrical interactions between the charged constituents (nuclei and electrons) of atoms and molecules:

- **Normal Reaction ($\mathbf{N}$ or $\mathbf{R}$):** The component of the mutual contact force perpendicular (normal) to the surfaces in contact.
- **Friction ($\mathbf{f}$):** The component of the mutual contact force parallel (tangential) to the surfaces in contact.
- **Fluid Contact Forces:** Upward buoyant force on a solid immersed in a fluid (equal to the weight of displaced fluid), viscous drag force in liquids, and air resistance.
- **Spring Force:** When an elastic spring is compressed or extended by a displacement x from its natural unstretched length, a restoring force $F = -kx$ is generated, where k is the force constant (spring constant) and the negative sign indicates that the restoring force opposes the displacement.
- **Tension in a String (T):** An inextensible string acts like a spring with a very high force constant. The restoring force in a taut string is called tension T. For a string of **negligible mass (massless string)**, the tension T has a constant magnitude throughout its length.

2.9 Friction: Static, Kinetic, and Rolling

Friction is the tangential component of the contact force between two surfaces that **opposes impending or actual relative motion** between the two surfaces.



Tutorial Diagram 3: (a) Self-adjusting static friction opposing impending motion, (b) kinetic friction opposing sliding, and (c) resolution of forces on an inclined plane.

1. Static Friction (f_s)

- **Self-Adjusting Nature:** Static friction does not exist by itself. When no tangential force is applied, $f_s = 0$. As an applied force F tries to cause relative motion, static friction comes into play and adjusts itself ($f_s = F$) up to a maximum limit to oppose **impending motion** (motion that would take place under the applied force if friction were absent).
- **Limiting Static Friction:** The maximum possible value of static friction, $(f_s)_{\max}$, is independent of the area of contact and proportional to the normal reaction N:

$$(f_s)_{\max} = \mu_s N \implies f_s \leq \mu_s N$$

where μ_s is the **coefficient of static friction**, which depends only on the nature of the two surfaces in contact.

- **Angle of Repose ($\theta_{\max}$) on an Inclined Plane:** When a block of mass m rests on an incline of angle θ, equilibrium requires $f_s = mg\sin\theta$ and $N = mg\cos\theta$. If the angle θ is gradually increased until the block is just on the verge of sliding at $\theta = \theta_{\max}$, static friction reaches its limiting value $(f_s)_{\max} = \mu_s N$:

$$mg\sin\theta_{\max} = \mu_s mg\cos\theta_{\max} \implies \tan\theta_{\max} = \mu_s \quad \text{or} \quad \theta_{\max} = \tan^{-1}(\mu_s)$$

Notice that $\theta_{\max}$ is completely independent of the mass m of the block.

2. Kinetic (Sliding) Friction (f_k)

- Once the applied force exceeds $(f_s)_{\max}$, relative sliding begins and the frictional force drops slightly to a nearly constant value called **kinetic friction (f_k)**:

$$f_k = \mu_k N$$

where μ_k is the **coefficient of kinetic friction**.

- Like static friction, f_k is independent of the area of contact and is nearly independent of the relative velocity between the surfaces. Experimentally, $\mu_k < \mu_s$.
- When sliding under an applied force F, the body's acceleration is $a = \frac{F - f_k}{m}$. For uniform velocity, $F = f_k$. If the applied force is removed, the retardation is $a = -\frac{f_k}{m}$.

3. Rolling Friction

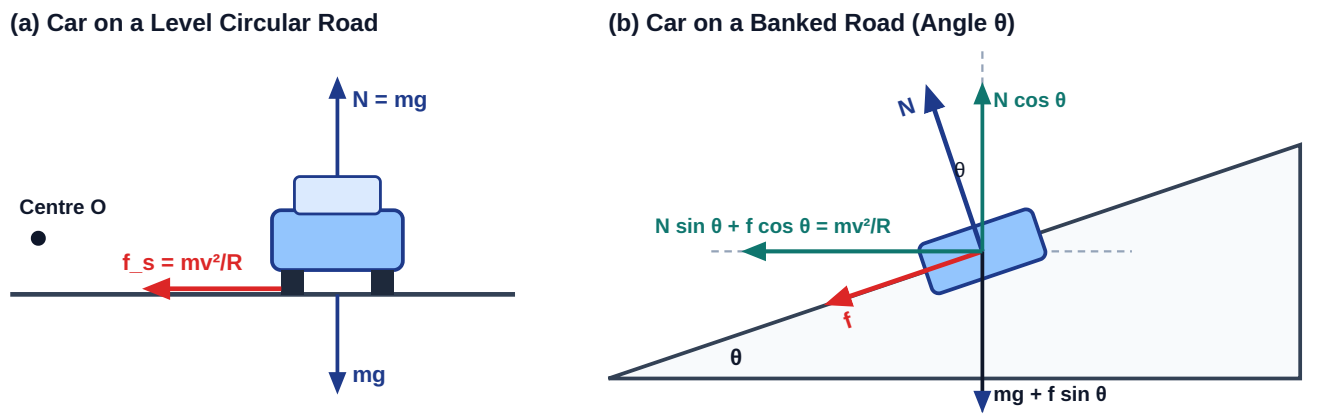
- In principle, a rigid ring or sphere rolling without slipping on an ideal horizontal plane has a single point of contact with zero relative velocity at any instant, so ideal static or kinetic friction is zero.
- In practice, the contacting surfaces momentarily deform slightly, creating a small finite area of contact whose tangential component opposes motion. This resistance is called **rolling friction**.
- For the same weight, rolling friction is **2 to 3 orders of magnitude smaller** than static or sliding friction.
- **Controlling Friction:** Because kinetic friction dissipates power as heat in machines, it is reduced using lubricants, **ball bearings** (which convert sliding friction into much smaller rolling friction between moving parts), or a thin **compressed air cushion** (as demonstrated by an inflated balloon attached to a plastic disc with a hole hovering over a flat surface). Conversely, friction is indispensable for walking, braking automobiles, and accelerating vehicles on roads.

2.10 Dynamics of Uniform Circular Motion

A body of mass m moving in a circle of radius R with uniform speed v has a radial acceleration $a_c = \frac{v^2}{R}$ directed toward the centre of the circle. By Newton's Second Law, the net radial inward force required to provide this acceleration is the **centripetal force**:

$$f_c = \frac{mv^2}{R}$$

Crucial Note: Centripetal force is **not a new kind of force**; it is simply the name given to whatever real material force (tension in a string, gravitational attraction on a planet, or friction on a car's tyres) provides the inward radial acceleration.



Tutorial Diagram 4: Free-body force resolution for a car taking a circular turn of radius R on (a) a level road and (b) a banked road.

Case 1: Motion of a Car on a Level Circular Road

Three forces act on the car: downward weight mg, upward normal reaction N, and inward static friction f (since the car tends to skid outward away from the centre of the circle):

- **Vertical Equilibrium:** $N - mg = 0 \implies N = mg.$
- **Horizontal Centripetal Equation:** Static friction provides the centripetal force:

$$f = \frac{mv^2}{R} \leq \mu_s N = \mu_s mg$$

- **Maximum Safe Speed on a Level Road:**

$$v^2 \leq \mu_s Rg \implies v_{\max} = \sqrt{\mu_s Rg}$$

  Note that $v_{\max}$ is independent of the mass m of the car.

Case 2: Motion of a Car on a Banked Circular Road

To reduce the reliance on friction and minimize tyre wear, circular roads are tilted (banked) upward at an angle θ with the horizontal. For a car moving near its maximum speed (impending outward skidding, so friction f acts **down the banked slope**):

- **Vertical Force Balance (zero vertical acceleration):**

$$N\cos\theta = mg + f\sin\theta \implies N(\cos\theta - \mu_s \sin\theta) = mg \quad (\text{at } f = \mu_s N)$$

- **Horizontal Centripetal Force Equation:**

$$N\sin\theta + f\cos\theta = \frac{mv^2}{R} \implies N(\sin\theta + \mu_s\cos\theta) = \frac{mv_{\max}^2}{R}$$

- **Maximum Permissible Speed on a Banked Road:** Substituting $N = \frac{mg}{\cos\theta - \mu_s \sin\theta}$ and dividing the numerator and denominator by $\cos\theta$:

$$v_{\max} = \left(Rg\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)^{1/2}$$

- **Optimum Speed (v_o) — Zero Friction Needed:** Setting $\mu_s = 0$ gives the optimum speed at which the horizontal component $N\sin\theta$ alone provides the exact required centripetal force:

$$v_o = (Rg\tan\theta)^{1/2} = \sqrt{Rg\tan\theta}$$

  Driving at $v = v_o$ requires no frictional force at all and causes minimal wear and tear of the tyres.
- **Direction of Friction when $v < v_o$:** If the car travels slower than v_o, it tends to slip *down* the incline, so static friction acts **up the slope**. A car can be parked stationary ($v = 0$) on the banked road only if $\tan\theta \leq \mu_s$.

3. Definitions, Equations & Physical Quantities

Quantity / Law	Symbol & Formula	SI Unit	Dimensions	Key Remarks
Linear Momentum	$\mathbf{p} = m\mathbf{v}$	$\mathrm{kg\,m\,s^{-1}}$ or $\mathrm{N\,s}$	$[\mathrm{M\,L\,T^{-1}}]$	Vector directed along velocity $\mathbf{v}$.
Force (Second Law)	$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = m\mathbf{a}$	N (newton)	$[\mathrm{M\,L\,T^{-2}}]$	Vector; $\mathbf{F}$ is net external force; local in space and time.
Impulse	$\mathbf{J} = \mathbf{F}\Delta t = \Delta\mathbf{p}$	$\mathrm{N\,s}$ or $\mathrm{kg\,m\,s^{-1}}$	$[\mathrm{M\,L\,T^{-1}}]$	Vector directed along $\Delta\mathbf{p}$; useful for short-duration impacts.
Static Friction	$f_s \leq (f_s)_{\max} = \mu_s N$	N	$[\mathrm{M\,L\,T^{-2}}]$	Self-adjusting; opposes impending relative motion; μ_s is dimensionless.
Kinetic Friction	$f_k = \mu_k N$	N	$[\mathrm{M\,L\,T^{-2}}]$	Opposes actual relative motion; experimentally $\mu_k < \mu_s$.
Spring Force	$F = -kx$	N (k in $\mathrm{N\,m^{-1}}$)	$[\mathrm{M\,L\,T^{-2}}]$	Restoring force opposite to elongation or compression x.
Centripetal Force	$f_c = \frac{mv^2}{R} = m\omega^2 R$	N	$[\mathrm{M\,L\,T^{-2}}]$	Directed radially inward toward centre; provided by material forces.

4. Systematic Procedure for Solving Problems in Mechanics

Typical mechanics problems involve an assembly of interconnected bodies (blocks, strings, pulleys, inclines, floors) exerting forces on one another while also experiencing gravity. To solve such problems without error, follow this five-step **Free-Body Diagram (FBD)** methodology:

1. **Draw the System Schematic:** Sketch the complete assembly of bodies, links, strings, pulleys, and supports.
2. **Select the System:** Choose a convenient part of the assembly (either a single body or a combined group of bodies moving together) as your *system*. Everything outside this choice constitutes the *environment*.
3. **Draw the Free-Body Diagram (FBD):** Isolate the chosen system and draw **all external forces acting ON the system** due to the environment and other agencies (gravity, normal reaction, tension, friction, applied forces).
   - *Golden Rule 1:* **Never include forces exerted BY the system on the environment.**
   - *Golden Rule 2:* **Never include internal forces** acting between parts that are both inside your chosen system.
   - *Golden Rule 3:* Do not draw $m\mathbf{a}$ as a force on the FBD; $\mathbf{a}$ is the effect of the net force.
4. **Specify Knowns and Unknowns:** Label known force magnitudes and directions (e.g., weight mg vertically downward, tension T away from the body along the string) and assign symbols to unknown forces.
5. **Apply Newton's Third Law Across Sub-Systems:** If you draw a second FBD for another body B in the assembly, ensure that if the force on A due to B was drawn as $\mathbf{F}$, then the force on B due to A is drawn as $-\mathbf{F}$ (equal and opposite).

5. Comprehensive Worked Examples

The following worked examples cover every major analytical pattern from the chapter.

Worked Example 1: Separation from an Accelerating Spaceship

Problem: An astronaut accidentally gets separated out of his small spaceship accelerating in interstellar space at a constant rate of $100\,\mathrm{m\,s^{-2}}$. What is the acceleration of the astronaut the instant after he is outside the spaceship? (Assume that there are no nearby stars to exert gravitational force on him.)

Solution:

Once the astronaut is outside the spaceship, there are no nearby stars to exert gravitational pull, and the small spaceship itself exerts negligible gravitational attraction on him. Since Newton's Second Law is a *local relation* (acceleration here and now is determined solely by the force acting here and now, with no memory of past acceleration), the net external force acting on the astronaut the instant he leaves the spaceship is zero ($\mathbf{F}_{\mathrm{net}} = \mathbf{0}$).

By Newton's First Law of Motion, the acceleration of the astronaut is **zero ($0\,\mathrm{m\,s^{-2}}$)**. (He continues to move with whatever constant velocity he had at the exact instant of separation).

Worked Example 2: Average Resistive Force on a Bullet & Equation of Motion

Problem: (a) A bullet of mass $0.04\,\text{kg}$ moving with a speed of $90\,\text{m s}^{-1}$ enters a heavy wooden block and is stopped after a distance of $60\,\text{cm}$. What is the average resistive force exerted by the block on the bullet?
(b) The motion of a particle of mass m is described by $y = ut + \frac{1}{2}gt^2$. Find the force acting on the particle.

Solution:

(a) Given $m = 0.04\,\text{kg}$, initial speed $u = 90\,\text{m s}^{-1}$, final speed $v = 0$, and stopping distance $s = 60\,\text{cm} = 0.6\,\text{m}$. Assuming uniform retardation a:

$$v^2 - u^2 = 2as \implies a = \frac{-u^2}{2s} = \frac{-90 \times 90}{2 \times 0.6} = -6750\,\text{m s}^{-2}$$

By Newton's Second Law, the average retarding force magnitude is:

$$F = m|a| = 0.04\,\text{kg} \times 6750\,\text{m s}^{-2} = 270\,\text{N}$$

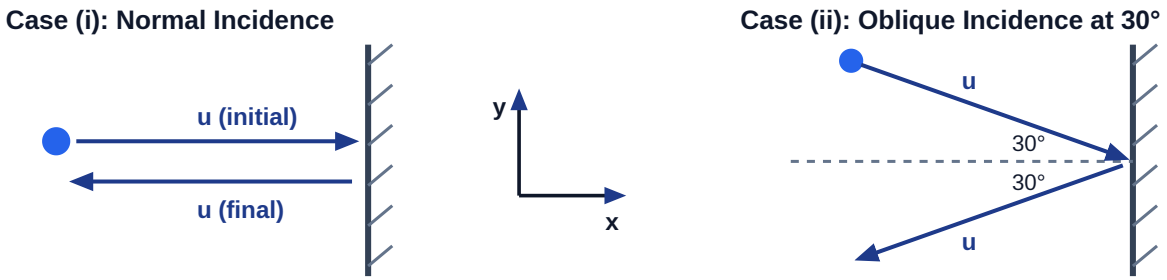
(b) Differentiating the position coordinate $y = ut + \frac{1}{2}gt^2$ with respect to time t:

$$v = \frac{\mathrm{d}y}{\mathrm{d}t} = u + gt \implies a = \frac{\mathrm{d}v}{\mathrm{d}t} = g$$

By Newton's Second Law, the force acting on the particle is $F = ma = mg$ (motion under gravity along the y-direction).

Worked Example 3: Impulse in Linear Rebound and Oblique Wall Collisions

Problem: (a) A batsman hits back a ball of mass $0.15\,\mathrm{kg}$ straight in the direction of the bowler without changing its initial speed of $12\,\mathrm{m\,s^{-1}}$. Determine the impulse imparted to the ball.
(b) Two identical billiard balls of mass m strike a rigid vertical wall (parallel to the y-axis) with the same speed u and rebound without any change in speed. In Case (i), the ball strikes the wall normally along the $+x$-axis and rebounds along the $-x$-axis. In Case (ii), the ball strikes the wall at an angle of 30° to the normal (in the $+x, -y$ direction) and reflects symmetrically at 30° to the normal (in the $-x, -y$ direction). Find the direction of the force on the wall due to each ball and the ratio of the magnitudes of impulses imparted to the balls by the wall.



Tutorial Diagram 5: Normal and oblique collisions of identical billiard balls with a rigid vertical wall.

Solution:

(a) Taking the direction from batsman to bowler as positive, initial momentum $p_i = -0.15 \times 12 = -1.8\,\mathrm{N\,s}$ and final momentum $p_f = +0.15 \times 12 = +1.8\,\mathrm{N\,s}$. Therefore:

$$\text{Impulse} = \Delta p = p_f - p_i = 1.8 - (-1.8) = 3.6\,\mathrm{N\,s} \quad \text{(directed from batsman to bowler)}$$

(b) Case (i): $(p_x)_i = +mu,\ (p_y)_i = 0$ and $(p_x)_f = -mu,\ (p_y)_f = 0$. Thus, the impulse imparted to the ball by the wall has components $J_x = -2mu$ and $J_y = 0$.

Case (ii): $(p_x)_i = +mu\cos 30^\circ,\ (p_y)_i = -mu\sin 30^\circ$ and $(p_x)_f = -mu\cos 30^\circ,\ (p_y)_f = -mu\sin 30^\circ$. Notice that p_x reverses sign while p_y remains unchanged! Thus, the impulse on the ball has components:

$$J_x = -2mu\cos 30^\circ, \qquad J_y = 0$$

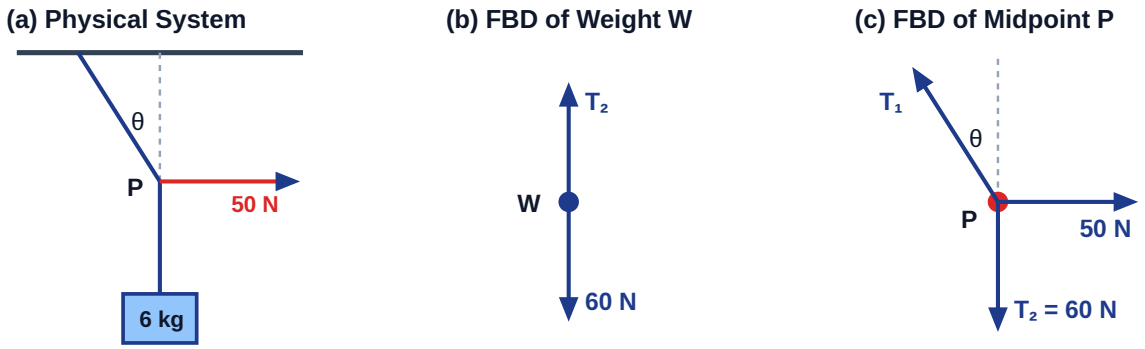
Conclusions:

1. In *both* cases, the impulse (and force) on the ball due to the wall is directed along the negative x-axis (normal to the wall). By Newton's Third Law, **the force on the wall due to the ball is normal to the wall along the positive x-axis in both Case (i) and Case (ii)**.
2. The ratio of the magnitudes of impulses imparted in Case (i) and Case (ii) is:

$$\frac{|J_{(\mathrm{i})}|}{|J_{(\mathrm{ii})}|} = \frac{2mu}{2mu\cos 30^\circ} = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}} \approx 1.2$$

Worked Example 4: Equilibrium of a Suspended Mass Pulled Horizontally

Problem: A mass of $6\,\mathrm{kg}$ is suspended by a massless rope of length $2\,\mathrm{m}$ from the ceiling. A horizontal force of $50\,\mathrm{N}$ is applied at the midpoint P of the rope. What is the angle θ the upper half of the rope makes with the vertical in equilibrium? (Take $g = 10\,\mathrm{m\,s^{-2}}$).



Tutorial Diagram 6: Free-body diagrams of the suspended weight W and the rope midpoint P.

Solution:

1. **Equilibrium of the $6\,\mathrm{kg}$ mass W:** The tension T_2 in the lower vertical segment balances the weight:

$$T_2 = mg = 6 \times 10 = 60\,\mathrm{N}$$

2. **Equilibrium of midpoint P:** Point P is in equilibrium under three concurrent forces: upper tension T_1 at angle θ to the vertical, downward tension $T_2 = 60\,\mathrm{N}$, and the horizontal force of $50\,\mathrm{N}$. Resolving forces horizontally and vertically:

$$T_1 \sin\theta = 50\,\mathrm{N} \quad \text{and} \quad T_1 \cos\theta = T_2 = 60\,\mathrm{N}$$

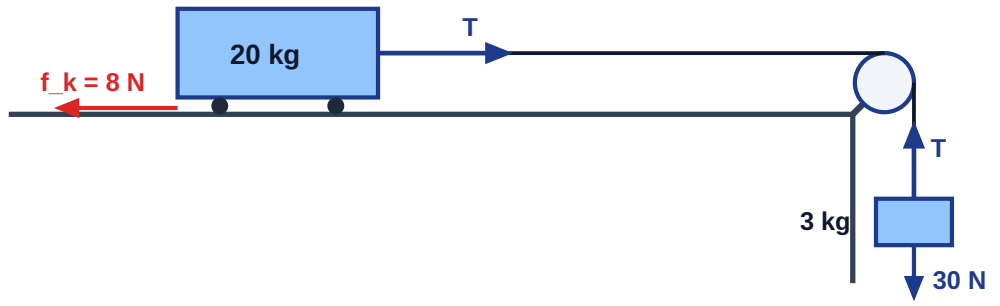
Dividing the two equations:

$$\tan\theta = \frac{50}{60} = \frac{5}{6} \implies \theta = \tan^{-1}\left(\frac{5}{6}\right) \approx 40^\circ$$

Note: Because the rope is massless, the angle θ depends neither on the total length of the rope nor on the point where the horizontal force is applied.

Worked Example 5: Friction on a Train Floor, Inclined Plane & Pulley-Trolley System

Problem: (a) Determine the maximum acceleration of a train in which a box lying on its floor will remain stationary relative to the train, given $\mu_s = 0.15$ and $g = 10\,\mathrm{m\,s^{-2}}$.
(b) A mass of $4\,\mathrm{kg}$ rests on a plane that is gradually inclined until at $\theta = 15^\circ$ with the horizontal, the mass just begins to slide. Find μ_s between the block and the surface.
(c) A $20\,\mathrm{kg}$ trolley resting on a horizontal surface ($\mu_k = 0.04$) is connected by a light inextensible string passing over a smooth pulley to a hanging $3\,\mathrm{kg}$ block. Find the acceleration of the system and the tension in the string ($g = 10\,\mathrm{m\,s^{-2}}$).



Tutorial Diagram 7: Trolley and hanging block connected over a smooth pulley.

Solution:

(a) Static friction provides the horizontal acceleration a of the box: $ma = f_s \le \mu_s N = \mu_s mg$. Thus:

$$a \le \mu_s g \implies a_{\max} = \mu_s g = 0.15 \times 10 = 1.5\,\mathrm{m\,s^{-2}}$$

(b) At the angle of repose $\theta_{\max} = 15^\circ$, limiting static friction balances the component of weight down the incline:

$$\mu_s = \tan\theta_{\max} = \tan 15^\circ = 0.27$$

(c) Because the string is inextensible and the pulley is smooth, both the $3\,\mathrm{kg}$ hanging block and the $20\,\mathrm{kg}$ trolley move with the same acceleration magnitude a.
• Equation of motion for the $3\,\mathrm{kg}$ hanging block: $30 - T = 3a$
• Normal force on the $20\,\mathrm{kg}$ trolley: $N = 20 \times 10 = 200\,\mathrm{N}$, so kinetic friction $f_k = \mu_k N = 0.04 \times 200 = 8\,\mathrm{N}$.
• Equation of motion for the $20\,\mathrm{kg}$ trolley: $T - f_k = 20a \implies T - 8 = 20a$.
Adding the two equations gives:

$$30 - 8 = 23a \implies a = \frac{22}{23}\,\mathrm{m\,s^{-2}} \approx 0.96\,\mathrm{m\,s^{-2}}$$

Substituting a back gives the tension: $T = 8 + 20 \times 0.96 = 27.1\,\mathrm{N}$.

Worked Example 6: Circular Turn on Level and Banked Roads

Problem: (a) A cyclist speeding at $18\,\text{km/h}$ on a level road takes a sharp circular turn of radius $3\,\text{m}$ without reducing speed. If $\mu_s = 0.1$ and $g = 9.8\,\text{m s}^{-2}$, will the cyclist slip?
(b) A circular racetrack of radius $300\,\text{m}$ is banked at an angle of 15°. If $\mu_s = 0.2$ between the wheels of a race-car and the road ($g = 9.8\,\text{m s}^{-2}$), find (i) the optimum speed to avoid wear and tear on its tyres, and (ii) the maximum permissible speed to avoid slipping.

Solution:

(a) Speed of cyclist $v = 18\,\text{km/h} = 18 \times \frac{5}{18} = 5\,\text{m s}^{-1}$, so $v^2 = 25\,\text{m}^2\,\text{s}^{-2}$.
Condition for not slipping on an unbanked level road is $v^2 \le \mu_s R g$. Here:

$$\mu_s R g = 0.1 \times 3 \times 9.8 = 2.94\,\text{m}^2\,\text{s}^{-2}$$

Since $v^2 = 25 > 2.94$, the condition is violated and **the cyclist will slip**.

(b) Given $R = 300\,\text{m}$, $\theta = 15^\circ$ ($\tan 15^\circ \approx 0.268$), $\mu_s = 0.2$, and $g = 9.8\,\text{m s}^{-2}$:
(i) **Optimum speed (v_o):**

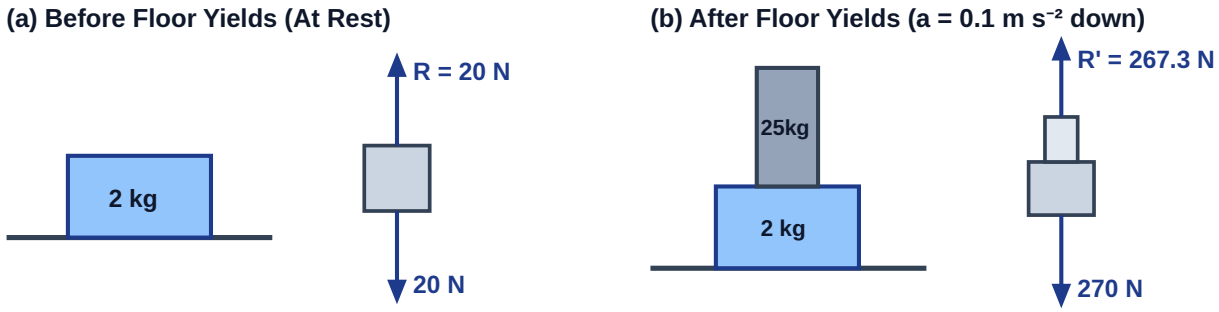
$$v_o = (Rg\tan\theta)^{1/2} = (300 \times 9.8 \times \tan 15^\circ)^{1/2} = 28.1\,\text{m s}^{-1}$$

(ii) **Maximum permissible speed ($v_{\max}$):**

$$v_{\max} = \left(Rg\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)^{1/2} = \left(300 \times 9.8 \times \frac{0.2 + 0.268}{1 - 0.2 \times 0.268}\right)^{1/2} = 38.1\,\text{m s}^{-1}$$

Worked Example 7: Action on a Yielding Floor & Action–Reaction Pairs

Problem: A wooden block of mass $2\,\text{kg}$ rests on a soft horizontal floor. When an iron cylinder of mass $25\,\text{kg}$ is placed on top of the block, the floor yields steadily and the block and the cylinder together go down with an acceleration of $0.1\,\text{m s}^{-2}$. What is the action of the block on the floor (a) before and (b) after the floor yields? (Take $g = 10\,\text{m s}^{-2}$). Identify the action–reaction pairs in the problem.



Tutorial Diagram 8: Free-body diagrams of (a) the 2 kg block at rest and (b) the combined (block + cylinder) system accelerating downward.

Solution:

(a) Before the floor yields: The $2\,\text{kg}$ block is at rest. Its FBD has two external forces: downward gravity $mg = 2 \times 10 = 20\,\text{N}$ and upward normal reaction R from the floor. By the First Law, $R = 20\,\text{N}$. By the Third Law, the action (force) of the block on the floor is $20\,\text{N}$ **directed vertically downwards**.

(b) After the floor yields: Choose the combined system $(\text{block} + \text{cylinder})$ of total mass $M = 2 + 25 = 27\,\text{kg}$, which accelerates downward at $a = 0.1\,\text{m s}^{-2}$. Its FBD does not include internal forces between the block and cylinder; the external forces are downward gravity $Mg = 27 \times 10 = 270\,\text{N}$ and upward normal reaction R' by the floor. By the Second Law:

$$270 - R' = 27 \times 0.1 = 2.7\,\text{N} \implies R' = 267.3\,\text{N}$$

By the Third Law, the action of the system (through the block) on the floor is $267.3\,\text{N}$ **vertically downwards**.

Action–Reaction Pairs:

• Pair 1: Gravitational pull of the Earth on the system ($20\,\text{N}$ in (a), $270\,\text{N}$ in (b)) downward $\leftrightarrow$ Gravitational pull of the system on the Earth upward.

• Pair 2: Downward contact force (action) of the block on the floor $\leftrightarrow$ Upward normal force R (or R') of the floor on the block.

• Pair 3 (in case b): Downward contact force of the cylinder on the block $\leftrightarrow$ Upward contact force of the block on the cylinder.

Warning: Weight mg and normal reaction R act on the **same body** and are **NEVER** an action–reaction pair!

6. NEET UG Practice Questions with Complete Worked Solutions

The following original practice questions are strictly aligned with the NEET UG Physics examination standard and cover all chapter concepts and problem types. Unless specified otherwise, take $g = 10\,\text{m s}^{-2}$.

Practice Question 1 [Conceptual / Multi-Situation — Newton's First Law]

Determine the magnitude of the net external force acting on each of the following five objects:

I. A drop of rain falling vertically down with a constant terminal speed.

II. A cork of mass $10\,\mathrm{g}$ floating stationary on the surface of water.

III. A kite skillfully held stationary in the sky against a blowing wind.

IV. A car moving with a constant velocity of $30\,\mathrm{km/h}$ on a rough horizontal road.

V. A high-speed electron in deep space far from all material objects and free of electric and magnetic fields.

(1) Zero in all five cases (I, II, III, IV, and V)

(2) Zero only in cases II, III, and V; non-zero in I and IV

(3) $0.1\,\mathrm{N}$ in case II; zero in all other cases

(4) Zero only in case V; non-zero in I, II, III, and IV

Correct Answer: (1)

Worked Solution:

By Newton's First Law of Motion, any body at rest ($\mathbf{v} = \mathbf{0}$) or moving with constant velocity ($\mathbf{a} = \mathbf{0}$) experiences **zero net external force ($\mathbf{F}_{\text{net}} = \mathbf{0}$)**:

• (I) Constant speed straight down $\implies \mathbf{a} = \mathbf{0} \implies F_{\text{net}} = 0$ (weight is balanced by upward buoyancy and viscous air drag).

• (II) Floating cork at rest $\implies \mathbf{a} = \mathbf{0} \implies F_{\text{net}} = 0$ (weight mg is balanced by upward buoyant force of water).

• (III) Stationary kite $\implies \mathbf{a} = \mathbf{0} \implies F_{\text{net}} = 0$ (force of wind balances string tension and weight).

• (IV) Constant velocity of $30\,\mathrm{km/h} \implies \mathbf{a} = \mathbf{0} \implies F_{\text{net}} = 0$.

• (V) Electron free of all fields and distant from matter $\implies F_{\text{net}} = 0$.

Practice Question 2 [Single-Correct MCQ — Force on a Projectile]

A pebble of mass $0.05\,\mathrm{kg}$ is thrown vertically upwards. Neglecting air resistance and taking $g = 10\,\mathrm{m\,s^{-2}}$, what is the magnitude and direction of the net force on the pebble (i) during its upward motion, (ii) during its downward motion, and (iii) at the highest point where it is momentarily at rest? Also, what happens to the net force if the pebble were thrown at an angle of 45° with the horizontal?

(1) $0.5\,\mathrm{N}$ upward in (i), $0.5\,\mathrm{N}$ downward in (ii), $0\,\mathrm{N}$ in (iii); changes to $0.5/\sqrt{2}\,\mathrm{N}$ if thrown at 45°.

(2) $0.5\,\mathrm{N}$ vertically downward in all three cases (i), (ii), and (iii), and remains $0.5\,\mathrm{N}$ vertically downward even if thrown at 45°.

(3) $0.5\,\mathrm{N}$ vertically downward in (i) and (ii), but $0\,\mathrm{N}$ at the highest point (iii) because velocity is zero.

(4) $0.05\,\mathrm{N}$ vertically downward in all cases.

Correct Answer: (2)

Worked Solution:

Once the pebble is in flight (with air resistance neglected), the *only* external force acting on it at every point of its trajectory is the gravitational pull of the Earth (its weight):

$$F_{\text{net}} = mg = 0.05\,\mathrm{kg} \times 10\,\mathrm{m\,s^{-2}} = 0.5\,\mathrm{N} \quad (\text{directed vertically downwards})$$

At the highest point, even though $\mathbf{v} = \mathbf{0}$ momentarily, acceleration is still $\mathbf{g}$ downward and force is still $0.5\,\mathrm{N}$ downward. If thrown at 45° with the horizontal, the horizontal velocity component remains constant while the net force on the pebble remains $mg = 0.5\,\mathrm{N}$ vertically downwards throughout flight.

Practice Question 3 [Statement-Based MCQ — Local Nature of Newton's Second Law]

A stone of mass $0.1\,\mathrm{kg}$ is placed in a train. Neglecting air resistance and taking $g = 10\,\mathrm{m\,s^{-2}}$, consider the net force acting on the stone in the following four situations:

- **Case A:** Just after it is dropped from the window of a stationary train.
- **Case B:** Just after it is dropped from the window of a train running at a constant velocity of $36\,\mathrm{km/h}$.
- **Case C:** Just after it is dropped from the window of a train accelerating horizontally at $1\,\mathrm{m\,s^{-2}}$.
- **Case D:** Lying stationary on the floor of a train that is accelerating horizontally at $1\,\mathrm{m\,s^{-2}}$.

Which of the following gives the correct magnitude and direction of the net force in Cases A, B, C, and D respectively?

(1) $1.0\,\mathrm{N}$ downward; $1.0\,\mathrm{N}$ downward; $\sqrt{1.01}\,\mathrm{N}$ inclined; $0.1\,\mathrm{N}$ horizontal

(2) $1.0\,\mathrm{N}$ vertically downward in A, B, and C; $0.1\,\mathrm{N}$ horizontally along the train's acceleration in D

(3) $0\,\mathrm{N}$ in A; $1.0\,\mathrm{N}$ downward in B; $0.1\,\mathrm{N}$ horizontal in C and D

(4) $1.0\,\mathrm{N}$ vertically downward in all four cases A, B, C, and D

Correct Answer: (2)

Worked Solution:

• **Cases A, B, and C:** Because Newton's Second Law is a *local relation*, the moment the stone is released from the train window, it loses all contact with the train and carries no "memory" of the train's acceleration. With air resistance neglected, the only force acting on the airborne stone in A, B, and C is gravity:

$$F_{\mathrm{net}} = mg = 0.1\,\mathrm{kg} \times 10\,\mathrm{m\,s^{-2}} = 1.0\,\mathrm{N} \quad (\text{vertically downwards})$$

• **Case D:** When the stone lies at rest relative to the floor of the train accelerating horizontally at $a = 1\,\mathrm{m\,s^{-2}}$, its vertical weight $(1.0\,\mathrm{N})$ is cancelled by the normal reaction of the floor, while horizontal static friction provides the net force required for its horizontal acceleration:

$$F_{\mathrm{net}} = ma = 0.1\,\mathrm{kg} \times 1\,\mathrm{m\,s^{-2}} = 0.1\,\mathrm{N} \quad (\text{horizontally along the train's motion})$$

Practice Question 4 [Single-Correct MCQ — Truck & Dropped Stone Kinematics]

A truck starts from rest and accelerates uniformly horizontally at $2.0\,\mathrm{m\,s^{-2}}$. At $t = 10\,\mathrm{s}$, a stone is dropped by a person standing on the top of the truck ($6\,\mathrm{m}$ high from the ground). Neglecting air resistance and taking $g = 10\,\mathrm{m\,s^{-2}}$, what are the magnitudes of the (a) velocity and (b) acceleration of the stone at $t = 11\,\mathrm{s}$?

(1) Velocity = $20\,\mathrm{m\,s^{-1}}$, Acceleration = $10\,\mathrm{m\,s^{-2}}$

(2) Velocity = $10\sqrt{5}\,\mathrm{m\,s^{-1}} \approx 22.4\,\mathrm{m\,s^{-1}}$, Acceleration = $10\,\mathrm{m\,s^{-2}}$ vertically downwards

(3) Velocity = $22.4\,\mathrm{m\,s^{-1}}$, Acceleration = $\sqrt{104}\,\mathrm{m\,s^{-2}}$

(4) Velocity = $30\,\mathrm{m\,s^{-1}}$, Acceleration = $2.0\,\mathrm{m\,s^{-2}}$

Correct Answer: (2)

Worked Solution:
At the instant $t = 10\,\mathrm{s}$ when the stone is dropped, the horizontal velocity of the truck (and therefore the initial velocity of the stone) is:

$$v_x = u + a_{\text{truck}}t = 0 + 2.0 \times 10 = 20\,\mathrm{m\,s^{-1}} \quad (\text{horizontal})$$

Once dropped at $t = 10\,\mathrm{s}$, there is no horizontal force on the stone ($a_x = 0$), and the only acceleration is due to gravity vertically downwards: $a_y = g = 10\,\mathrm{m\,s^{-2}}$. Thus, at $t = 11\,\mathrm{s}$ (which is $\Delta t = 1\,\mathrm{s}$ after release):
• Horizontal velocity remains $v_x = 20\,\mathrm{m\,s^{-1}}$.
• Vertical downward velocity acquired in $\Delta t = 1\,\mathrm{s}$ is $v_y = g\Delta t = 10 \times 1 = 10\,\mathrm{m\,s^{-1}}$.
• Resultant speed of the stone at $t = 11\,\mathrm{s}$:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{20^2 + 10^2} = \sqrt{500} = 10\sqrt{5} \approx 22.4\,\mathrm{m\,s^{-1}}$$

(at an angle $\tan^{-1}(10/20) = \tan^{-1}(0.5) \approx 26.6^\circ$ below the horizontal), and its acceleration is $10\,\mathrm{m\,s^{-2}}$ vertically downwards.

Practice Question 5 [Single-Correct MCQ — Constant Force & Retardation]

(a) A constant retarding force of $50\,\mathrm{N}$ is applied to a body of mass $20\,\mathrm{kg}$ moving initially with a speed of $15\,\mathrm{m\,s^{-1}}$. How long does the body take to stop?
(b) A constant force acting on a body of mass $3.0\,\mathrm{kg}$ changes its speed from $2.0\,\mathrm{m\,s^{-1}}$ to $3.5\,\mathrm{m\,s^{-1}}$ in $25\,\mathrm{s}$ without changing its direction of motion. What is the magnitude and direction of the force?

(1) (a) $6.0\,\mathrm{s}$; (b) $0.18\,\mathrm{N}$ in the direction of motion

(2) (a) $12.0\,\mathrm{s}$; (b) $0.18\,\mathrm{N}$ opposite to motion

(3) (a) $6.0\,\mathrm{s}$; (b) $0.66\,\mathrm{N}$ in the direction of motion

(4) (a) $3.0\,\mathrm{s}$; (b) $0.18\,\mathrm{N}$ in the direction of motion

Correct Answer: (1)

Worked Solution:
(a) Acceleration $a = \frac{F}{m} = \frac{-50\,\mathrm{N}}{20\,\mathrm{kg}} = -2.5\,\mathrm{m\,s^{-2}}$. Using $v = u + at$ with $v = 0$ and $u = 15\,\mathrm{m\,s^{-1}}$:

$$0 = 15 - 2.5t \implies t = \frac{15}{2.5} = 6.0\,\mathrm{s}$$

(b) Acceleration $a = \frac{v-u}{t} = \frac{3.5-2.0}{25} = \frac{1.5}{25} = 0.06\,\mathrm{m\,s^{-2}}$. Force magnitude:

$$F = ma = 3.0\,\mathrm{kg} \times 0.06\,\mathrm{m\,s^{-2}} = 0.18\,\mathrm{N}$$

Since the speed increases along the unchanged line of motion, the force acts **in the direction of motion**.

Practice Question 6 [Single-Correct MCQ — Perpendicular Forces & Vehicle Braking]

(a) A body of mass $5\,\text{kg}$ is acted upon by two perpendicular forces of $8\,\text{N}$ and $6\,\text{N}$. Find the magnitude and direction of the acceleration of the body.
(b) The driver of a three-wheeler moving at $36\,\text{km/h}$ sees a child standing in the middle of the road and brings his vehicle to rest in $4.0\,\text{s}$ just in time to save the child. If the mass of the three-wheeler is $400\,\text{kg}$ and the mass of the driver is $65\,\text{kg}$, what is the average retarding force on the vehicle?

(1) (a) $2.8\,\text{m s}^{-2}$ at 45°; (b) $1000\,\text{N}$

(2) (a) $2\,\text{m s}^{-2}$ at $\tan^{-1}(3/4) \approx 37^\circ$ with the $8\,\text{N}$ force; (b) $1162.5\,\text{N}$

(3) (a) $2\,\text{m s}^{-2}$ at $\tan^{-1}(4/3) \approx 53^\circ$ with the $8\,\text{N}$ force; (b) $1162.5\,\text{N}$

(4) (a) $10\,\text{m s}^{-2}$ at 37° with the $8\,\text{N}$ force; (b) $4650\,\text{N}$

Correct Answer: (2)

Worked Solution:
(a) Resultant of the two perpendicular forces $F_1 = 8\,\text{N}$ and $F_2 = 6\,\text{N}$:

$$F = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = 10\,\text{N} \implies a = \frac{F}{m} = \frac{10\,\text{N}}{5\,\text{kg}} = 2\,\text{m s}^{-2}$$

If θ is the angle made by the resultant force (and acceleration) with the $8\,\text{N}$ force, then $\tan\theta = \frac{6}{8} = \frac{3}{4} = 0.75 \implies \theta \approx 37^\circ$.
(b) Initial speed $u = 36\,\text{km/h} = 10\,\text{m s}^{-1}$, final speed $v = 0$, $t = 4.0\,\text{s}$, and total mass $M = 400 + 65 = 465\,\text{kg}$.
Retardation $a = \frac{v-u}{t} = \frac{0-10}{4.0} = -2.5\,\text{m s}^{-2}$. Magnitude of average retarding force:

$$F = M|a| = 465\,\text{kg} \times 2.5\,\text{m s}^{-2} = 1162.5\,\text{N}$$

Practice Question 7 [Single-Correct MCQ — Rocket Blast Thrust]

A rocket with a lift-off mass of $20,000\,\text{kg}$ is blasted upwards with an initial upward acceleration of $5.0\,\text{m s}^{-2}$. Taking $g = 10\,\text{m s}^{-2}$, calculate the initial thrust (upward force) of the blast.

(1) $1.0 \times 10^5\,\text{N}$

(2) $2.0 \times 10^5\,\text{N}$

(3) $3.0 \times 10^5\,\text{N}$

(4) $4.0 \times 10^5\,\text{N}$

Correct Answer: (3)

Worked Solution:
Two external forces act on the rocket at lift-off: upward thrust T of the blast and downward weight mg. Applying Newton's Second Law in the upward direction:

$$T - mg = ma \implies T = m(g + a)$$

$$T = 20,000\,\text{kg} \times (10 + 5.0)\,\text{m s}^{-2} = 20,000 \times 15 = 3.0 \times 10^5\,\text{N}$$

Practice Question 8 [Single-Correct MCQ — Piecewise Force & Position Prediction]

A body of mass $0.40\,\text{kg}$ moving initially with a constant speed of $10\,\text{m}\,\text{s}^{-1}$ to the north is subjected to a constant force of $8.0\,\text{N}$ directed towards the south for $30\,\text{s}$. Take the instant the force is applied to be $t = 0$, the position of the body at that time to be $x = 0$ (with north as positive x), and predict its position at $t = -5\,\text{s}$, $t = 25\,\text{s}$, and $t = 100\,\text{s}$.

(1) $-50\,\text{m}$, $-6000\,\text{m}$, and $-50,000\,\text{m}$

(2) $-50\,\text{m}$, $-6000\,\text{m}$, and $-99,000\,\text{m}$

(3) $+50\,\text{m}$, $-6250\,\text{m}$, and $-50,000\,\text{m}$

(4) $-50\,\text{m}$, $-6500\,\text{m}$, and $-45,000\,\text{m}$

Correct Answer: (1)

Worked Solution:

Take north as positive x. Initial velocity at $t = 0$ is $u = +10\,\text{m}\,\text{s}^{-1}$.

• **At $t = -5\,\text{s}$ (before force is applied):** The body moves with uniform velocity $u = 10\,\text{m}\,\text{s}^{-1}$, so:

$$x(-5\,\text{s}) = ut = 10 \times (-5) = -50\,\text{m}$$

• **For $0 \le t \le 30\,\text{s}$ (while the $8.0\,\text{N}$ southward force acts):** Acceleration is:

$$a = \frac{F}{m} = \frac{-8.0\,\text{N}}{0.40\,\text{kg}} = -20\,\text{m}\,\text{s}^{-2}$$

At $t = 25\,\text{s}$ (which falls within the $30\,\text{s}$ interval):

$$x(25\,\text{s}) = ut + \frac{1}{2}at^2 = (10)(25) + \frac{1}{2}(-20)(25)^2 = 250 - 6250 = -6000\,\text{m} \quad (-6\,\text{km})$$

• **At $t = 100\,\text{s}$:** First find the position and velocity at $t = 30\,\text{s}$ when the force ceases:

$x(30\,\text{s}) = (10)(30) + \frac{1}{2}(-20)(30)^2 = 300 - 9000 = -8700\,\text{m}$

$v(30\,\text{s}) = u + at = 10 + (-20)(30) = -590\,\text{m}\,\text{s}^{-1}$

For the remaining $100 - 30 = 70\,\text{s}$, no force acts ($a = 0$), so the body moves at constant velocity $-590\,\text{m}\,\text{s}^{-1}$:

$x(100\,\text{s}) = x(30\,\text{s}) + v(30\,\text{s}) \times 70 = -8700 + (-590)(70) = -8700 - 41,300 = -50,000\,\text{m} \quad (-50\,\text{km})$

Practice Question 9 [Single-Correct MCQ — Apparent Weight in an Accelerating Lift]

A man of mass $70\,\text{kg}$ stands on a weighing scale in a lift. Taking $g = 10\,\text{m s}^{-2}$, what would be the readings on the scale (in kg) in the following four cases?
(a) Lift moving upwards with a uniform speed of $10\,\text{m s}^{-1}$.
(b) Lift moving downwards with a uniform acceleration of $5\,\text{m s}^{-2}$.
(c) Lift moving upwards with a uniform acceleration of $5\,\text{m s}^{-2}$.
(d) The lift mechanism fails and it hurtles down freely under gravity.

(1) (a) $70\,\text{kg}$, (b) $35\,\text{kg}$, (c) $105\,\text{kg}$, (d) $0\,\text{kg}$

(2) (a) $140\,\text{kg}$, (b) $35\,\text{kg}$, (c) $105\,\text{kg}$, (d) $70\,\text{kg}$

(3) (a) $70\,\text{kg}$, (b) $105\,\text{kg}$, (c) $35\,\text{kg}$, (d) $0\,\text{kg}$

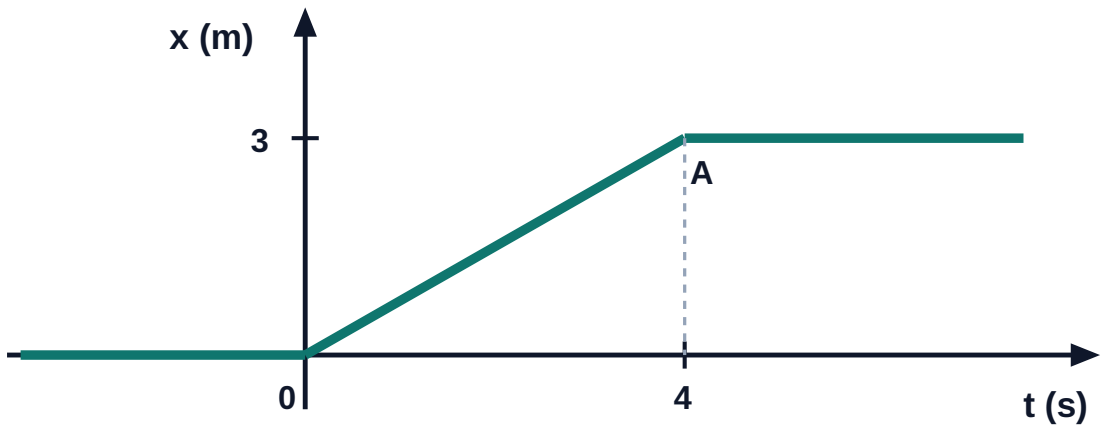
(4) (a) $0\,\text{kg}$, (b) $35\,\text{kg}$, (c) $105\,\text{kg}$, (d) $0\,\text{kg}$

Correct Answer: (1)

Worked Solution:
A weighing scale measures the normal reaction R (in newtons) and displays the apparent mass $m_{\text{app}} = R/g$ (in kg):
• **(a) Uniform speed** ($a = 0$): $R - mg = 0 \implies R = 70 \times 10 = 700\,\text{N} \implies \text{Reading} = 70\,\text{kg}$.
• **(b) Downward acceleration** ($a = 5\,\text{m s}^{-2}$): $mg - R = ma \implies R = m(g - a) = 70(10 - 5) = 350\,\text{N} \implies \text{Reading} = 35\,\text{kg}$.
• **(c) Upward acceleration** ($a = 5\,\text{m s}^{-2}$): $R - mg = ma \implies R = m(g + a) = 70(10 + 5) = 1050\,\text{N} \implies \text{Reading} = 105\,\text{kg}$.
• **(d) Free fall** ($a = g = 10\,\text{m s}^{-2}$ **downward**): $R = m(g - g) = 0\,\text{N} \implies \text{Reading} = 0\,\text{kg}$ (state of weightlessness).

Practice Question 10 [Graph-Based MCQ — Force & Impulse from Position-Time Graph]

The one-dimensional position-time (x-t) graph of a particle of mass $4\,\text{kg}$ is shown in the diagram below. What is (a) the force on the particle for $t < 0$, $0 < t < 4\,\text{s}$, and $t > 4\,\text{s}$, and (b) the impulse imparted to the particle at $t = 0$ and at $t = 4\,\text{s}$?



Tutorial Diagram 9: Position-time (x-t) graph for a particle of mass 4 kg.

(1) (a) Force = $0\,\text{N}$ in all three intervals; (b) Impulse at $t = 0$ is $+3\,\text{kg m s}^{-1}$, and at $t = 4\,\text{s}$ is $-3\,\text{kg m s}^{-1}$

(2) (a) Force = $0\,\text{N}$ for $t < 0$ and $t > 4\,\text{s}$, and $3\,\text{N}$ for \(0

(3) (a) Force = $0\,\text{N}$ in all three intervals; (b) Impulse at $t = 0$ is $+0.75\,\text{kg m s}^{-1}$, and at $t = 4\,\text{s}$ is $-0.75\,\text{kg m s}^{-1}$

(4) (a) Force = $12\,\text{N}$ for \(0

Correct Answer: (1)

Worked Solution:

(a) Force in the open intervals:

• For $t < 0$: $x = 0$ (particle at rest, $v_1 = 0,\ a = 0$) $\implies F = 0$.

• For $0 < t < 4\,\text{s}$: The x-t graph is a straight line with constant slope $v_2 = \frac{3-0}{4-0} = 0.75\,\text{m s}^{-1}$. Since velocity is constant, acceleration $a = 0 \implies F = 0$.

• For $t > 4\,\text{s}$: $x = 3\,\text{m}$ (particle at rest, $v_3 = 0,\ a = 0$) $\implies F = 0$.

(b) Impulse at the corners $t = 0$ and $t = 4\,\text{s}$:

• At $t = 0$: Velocity jumps from $v_1 = 0$ to $v_2 = 0.75\,\text{m s}^{-1}$. Impulse $= \Delta p = m(v_2 - v_1) = 4(0.75 - 0) = +3\,\text{kg m s}^{-1}$.

• At $t = 4\,\text{s}$: Velocity jumps from $v_2 = 0.75\,\text{m s}^{-1}$ to $v_3 = 0$. Impulse $= \Delta p = m(v_3 - v_2) = 4(0 - 0.75) = -3\,\text{kg m s}^{-1}$.

Practice Question 11 [Single-Correct MCQ — Connected Blocks & Atwood's Machine]

(a) Two bodies A and B of masses $10\,\text{kg}$ and $20\,\text{kg}$ respectively are kept on a smooth horizontal surface and tied to the ends of a light string. A horizontal force $F = 600\,\text{N}$ is applied along the direction of the string (i) to A (pulling B behind it), and (ii) to B (pulling A behind it). What is the tension in the string in each case?
(b) Two masses $8\,\text{kg}$ and $12\,\text{kg}$ are connected at the two ends of a light inextensible string that goes over a frictionless pulley. Find the acceleration of the masses and the tension in the string when released ($g = 10\,\text{m s}^{-2}$).

(1) (a) (i) $300\,\text{N}$, (ii) $300\,\text{N}$; (b) $a = 2\,\text{m s}^{-2},\ T = 96\,\text{N}$

(2) (a) (i) $400\,\text{N}$, (ii) $200\,\text{N}$; (b) $a = 2\,\text{m s}^{-2},\ T = 96\,\text{N}$

(3) (a) (i) $200\,\text{N}$, (ii) $400\,\text{N}$; (b) $a = 4\,\text{m s}^{-2},\ T = 80\,\text{N}$

(4) (a) (i) $400\,\text{N}$, (ii) $200\,\text{N}$; (b) $a = 2.5\,\text{m s}^{-2},\ T = 100\,\text{N}$

Correct Answer: (2)

Worked Solution:
(a) Connected blocks on smooth surface: Total mass $M = m_A + m_B = 10 + 20 = 30\,\text{kg}$. In both cases, common acceleration is $a = \frac{F}{M} = \frac{600}{30} = 20\,\text{m s}^{-2}$.
• Case (i): $F = 600\,\text{N}$ pulls A ($10\,\text{kg}$), and tension T_1 pulls the trailing mass B ($20\,\text{kg}$):

$$T_1 = m_B a = 20\,\text{kg} \times 20\,\text{m s}^{-2} = 400\,\text{N}$$

• Case (ii): $F = 600\,\text{N}$ pulls B ($20\,\text{kg}$), and tension T_2 pulls the trailing mass A ($10\,\text{kg}$):

$$T_2 = m_A a = 10\,\text{kg} \times 20\,\text{m s}^{-2} = 200\,\text{N}$$

(b) Atwood's pulley system ($m_1 = 8\,\text{kg},\ m_2 = 12\,\text{kg}$):
Equations of motion are $12g - T = 12a$ and $T - 8g = 8a$. Adding gives:

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(12 - 8) \times 10}{12 + 8} = \frac{40}{20} = 2\,\text{m s}^{-2}$$

$$T = 8(g + a) = 8(10 + 2) = 96\,\text{N}$$

Practice Question 12 [Single-Correct MCQ — Momentum Conservation & Vector Impulse]

(a) Two billiard balls each of mass $0.05\,\text{kg}$ moving in opposite directions with speed $6\,\text{m}\,\text{s}^{-1}$ collide head-on and rebound with the same speed. What is the magnitude of the impulse imparted to each ball due to the other?
(b) A shell of mass $0.020\,\text{kg}$ is fired by a gun of mass $100\,\text{kg}$. If the muzzle speed of the shell is $80\,\text{m}\,\text{s}^{-1}$, what is the recoil speed of the gun?
(c) A batsman deflects a ball of mass $0.15\,\text{kg}$ by an angle of 45° without changing its initial speed of $54\,\text{km/h}$. What is the magnitude of the impulse imparted to the ball?

(1) (a) $0.6\,\text{kg}\,\text{m}\,\text{s}^{-1}$; (b) $0.016\,\text{m}\,\text{s}^{-1}$; (c) $4.16\,\text{kg}\,\text{m}\,\text{s}^{-1}$

(2) (a) $0.3\,\text{kg}\,\text{m}\,\text{s}^{-1}$; (b) $0.016\,\text{m}\,\text{s}^{-1}$; (c) $2.25\,\text{kg}\,\text{m}\,\text{s}^{-1}$

(3) (a) $0.6\,\text{kg}\,\text{m}\,\text{s}^{-1}$; (b) $0.16\,\text{m}\,\text{s}^{-1}$; (c) $3.18\,\text{kg}\,\text{m}\,\text{s}^{-1}$

(4) (a) $0\,\text{kg}\,\text{m}\,\text{s}^{-1}$; (b) $0.016\,\text{m}\,\text{s}^{-1}$; (c) $4.16\,\text{kg}\,\text{m}\,\text{s}^{-1}$

Correct Answer: (1)

Worked Solution:

(a) For either ball, initial momentum is $p_i = +0.05 \times 6 = +0.3\,\text{kg}\,\text{m}\,\text{s}^{-1}$ and final momentum after rebounding is $p_f = -0.05 \times 6 = -0.3\,\text{kg}\,\text{m}\,\text{s}^{-1}$. Magnitude of impulse on each ball:

$$|J| = |p_f - p_i| = |-0.3 - 0.3| = 0.6\,\text{kg}\,\text{m}\,\text{s}^{-1} \quad (\text{opposite to its initial velocity})$$

(b) By conservation of linear momentum ($p_{\text{initial}} = 0$):

$$M_{\text{gun}} V = m_{\text{shell}} v \implies V = \frac{0.020\,\text{kg} \times 80\,\text{m}\,\text{s}^{-1}}{100\,\text{kg}} = 0.016\,\text{m}\,\text{s}^{-1} \quad (1.6\,\text{cm}\,\text{s}^{-1})$$

(c) Speed of ball $u = 54\,\text{km/h} = 54 \times \frac{5}{18} = 15\,\text{m}\,\text{s}^{-1}$. Since the ball is deflected by $\theta = 45^\circ$ without change in speed (meaning the velocity vector reverses and deflects so the angle between $\mathbf{p}_i$ and $-\mathbf{p}_f$ is 45°, i.e. each makes 22.5° with the symmetry bisector):

$$|J| = |\Delta \mathbf{p}| = 2mu\cos(22.5^\circ) = 2 \times 0.15 \times 15 \times 0.9239 \approx 4.16\,\text{kg}\,\text{m}\,\text{s}^{-1}$$

Practice Question 13 [Single-Correct MCQ — Horizontal Circular Motion & String Breakage]

One end of a string of length $l = 1.5\,\text{m}$ is connected to a stone of mass $m = 0.25\,\text{kg}$ and the other end to a small peg on a smooth horizontal table. The stone is whirled in a horizontal circle at a speed of $40\,\text{rev/min}$.
(i) What is the expression for the net force on the stone directed towards the centre (in terms of tension T)?
(ii) What is the numerical value of the tension T in the string, and what is the maximum speed $v_{\max}$ with which the stone can be whirled if the string can withstand a maximum tension of $200\,\text{N}$?
(iii) If the speed is increased beyond $v_{\max}$ and the string breaks suddenly, what describes the trajectory of the stone on the horizontal table after the string breaks?

(1) (i) $T - \frac{mv^2}{l}$; (ii) $T = 6.6\,\text{N},\ v_{\max} = 34.6\,\text{m s}^{-1}$; (iii) moves radially outwards

(2) (i) T; (ii) $T = 6.6\,\text{N},\ v_{\max} = 34.6\,\text{m s}^{-1}$; (iii) flies off tangentially from the instant the string breaks

(3) (i) T; (ii) $T = 13.2\,\text{N},\ v_{\max} = 20.0\,\text{m s}^{-1}$; (iii) flies off tangentially

(4) (i) 0; (ii) $T = 6.6\,\text{N},\ v_{\max} = 34.6\,\text{m s}^{-1}$; (iii) flies off at an angle to the tangent

Correct Answer: (2)

Worked Solution:

• **(i)** The net inward radial force on the particle *is* the tension T in the string (which equals $\frac{mv^2}{l}$; we must never subtract $\frac{mv^2}{l}$ from T in an inertial frame!).

• **(ii)** Angular speed $\omega = \frac{40 \times 2\pi}{60} = \frac{4\pi}{3}\,\text{rad s}^{-1}$. Tension in the string is:

$$T = m\omega^2 R = 0.25 \times \left(\frac{4\pi}{3}\right)^2 \times 1.5 \approx 6.6\,\text{N}$$

For $T_{\max} = 200\,\text{N}$:

$$\frac{mv_{\max}^2}{R} = T_{\max} \implies v_{\max} = \sqrt{\frac{T_{\max} R}{m}} = \sqrt{\frac{200 \times 1.5}{0.25}} = \sqrt{1200} \approx 34.6\,\text{m s}^{-1}$$

• **(iii)** The moment the string breaks, the horizontal centripetal force becomes zero. By Newton's First Law, the stone continues in a straight line along the direction of its instantaneous velocity—so **it flies off tangentially from the instant the string breaks**.

Practice Question 14 [Conceptual / Assertion–Reason Style — Oscillating Bob & Everyday Mechanics]

Consider the following two sets of mechanical situations:

Part I: A bob of mass $0.1\,\mathrm{kg}$ hung from the ceiling of a room by a string $2\,\mathrm{m}$ long is set into oscillation. The speed of the bob at its mean position is $1\,\mathrm{m\,s^{-1}}$. What is the trajectory of the bob if the string is cut (a) at one of its extreme positions, and (b) at its mean position?

Part II: Evaluate the physical reasons for the following statements:

• Statement P: A horse cannot pull a cart and run in empty space because there is no external agency (ground) to exert a forward reaction force on the horse.

• Statement Q: It is easier to pull a lawn mower than to push it at an angle to the horizontal because pulling has a vertical upward component that reduces the normal reaction N (and hence reduces friction), whereas pushing has a downward vertical component that increases N and friction.

• Statement R: If a stationary nucleus disintegrates into two smaller nuclei, the products must move in opposite directions so that their momenta cancel ($\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{0}$).

(1) Part I: (a) Parabolic path, (b) Vertically downwards; Part II: Only P and R are correct

(2) Part I: (a) Vertically downwards, (b) Parabolic path; Part II: All three statements P, Q, and R are physically correct

(3) Part I: (a) Tangential upward path, (b) Horizontal straight line; Part II: All three statements P, Q, and R are correct

(4) Part I: (a) Vertically downwards, (b) Vertically downwards; Part II: Only Q and R are correct

Correct Answer: (2)

Worked Solution:

• **Part I (a) At an extreme position:** The instantaneous velocity of the oscillating bob is zero ($\mathbf{v} = \mathbf{0}$). When the string is cut, only downward gravity acts on it, so the bob **falls vertically downwards**.

• **Part I (b) At the mean position:** The bob has a horizontal velocity of $1\,\mathrm{m\,s^{-1}}$. When the string is cut, it becomes a horizontal projectile under gravity and follows a **parabolic trajectory**.

• **Part II:** Statements P, Q, and R are all completely accurate applications of Newton's Third Law, friction ($f_k = \mu_k N$), and conservation of linear momentum ($\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{0} \implies \mathbf{p}_1 = -\mathbf{p}_2$).

7. Common Pitfalls & Points to Ponder (NEET Exam Traps)

1. Force is NOT always in the direction of velocity ($\mathbf{v}$): Depending on the physical situation, $\mathbf{F}$ may be parallel to $\mathbf{v}$ (speeding up), antiparallel to $\mathbf{v}$ (slowing down), perpendicular to $\mathbf{v}$ (uniform circular motion), or at an arbitrary angle (projectile motion). However, $\mathbf{F}$ **is ALWAYS parallel to the instantaneous acceleration** $\mathbf{a}$.

2. Zero instantaneous velocity ($\mathbf{v} = \mathbf{0}$) does NOT imply zero force or zero acceleration: When a ball thrown vertically upward reaches its highest point, $\mathbf{v} = \mathbf{0}$ at that instant, but the force continues to be its downward weight mg and its acceleration remains g downward.

3. Force is NOT "carried" from past motion: Force at a given instant is determined solely by the interactions at the body's location at that instant. The moment a stone is released outside an accelerating train, its horizontal force and horizontal acceleration immediately drop to zero (neglecting air resistance).

4. Do NOT treat $m\mathbf{a}$ or "centripetal force" as additional forces in an FBD: In $\mathbf{F} = m\mathbf{a}$, $\mathbf{F}$ is the vector sum of real material external forces (gravity, tension, normal reaction, friction), whereas $\mathbf{a}$ is the *effect*. Similarly, centripetal force $\frac{mv^2}{R}$ is simply the name given to the inward radial resultant of real material forces.

5. Never blindly substitute $f_s = \mu_s N$: Static friction is a self-adjusting force satisfying $f_s \leq \mu_s N$. Set $f_s = \mu_s N$ *only* when the body is explicitly on the verge of slipping (limiting equilibrium).

6. $mg = R$ is NOT a universal law and has NO connection with Newton's Third Law: For a body on a horizontal surface, $mg = R$ holds only when vertical acceleration is zero. In an accelerating lift, $R \neq mg$. Furthermore, mg and R act on the *same body*, whereas an action–reaction pair always acts on *two different bodies*.

7. Objective force vs. Subjective "feeling of force": On a rotating merry-go-round, every part of our body experiences an *inward* centripetal force, even though we subjectively feel "pushed outward" in the direction of impending motion. Also, animate objects (humans, horses) obey the exact same laws of motion as inanimate objects: an external force (friction from the ground) is essential for walking or running.

8. Concise Final Revision Recap

- **First Law (Law of Inertia):** $\mathbf{F}_{\text{net}} = \mathbf{0} \iff \mathbf{a} = \mathbf{0}$ (rest and uniform linear motion are dynamically equivalent).
- **Second Law & Impulse:** $\mathbf{F}_{\text{net}} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = m\mathbf{a}$; $\text{Impulse } \mathbf{J} = \mathbf{F}\Delta t = \Delta\mathbf{p}$.
- **Third Law:** $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$ (simultaneous mutual forces acting on different bodies; internal forces in a system cancel in pairs).
- **Conservation of Momentum:** If $\mathbf{F}_{\text{ext}} = \mathbf{0}$, then $\sum \mathbf{p}_i = \sum \mathbf{p}_f = \text{constant}$ (valid for all elastic and inelastic collisions, recoil, and disintegrations).
- **Friction Laws:** $f_s \leq \mu_s N$; angle of repose $\tan\theta_{\max} = \mu_s$; $f_k = \mu_k N$ ($\mu_k < \mu_s$); rolling friction $\ll$ sliding friction.
- **Circular Turn Formulas:**
  - Level road maximum speed: $v_{\max} = \sqrt{\mu_s R g}$
  - Banked road optimum speed ($\mu_s = 0$): $v_o = \sqrt{Rg\tan\theta}$
  - Banked road maximum safe speed: $v_{\max} = \sqrt{Rg\left(\frac{\mu_s + \tan\theta}{1 - \mu_s\tan\theta}\right)}$